

AI-Based Women Safety Smart Slipper

**INNOVATIVE
PROJECT
April/May
2026**

INNOVATIVE / MULTI-DISCIPLINARY PROJECT REPORT
SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF THE
DEGREE OF **BACHELOR OF TECHNOLOGY**
IN **ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**
OF THE ANNA UNIVERSITY



**PROJECT
WORK**

Submitted by
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**BATCH
2023 – 2027**

Under the Guidance of
Ms.R.Geetha Rajakumari
Assistant Professor
Artificial Intelligence And Data Science



Sri Eshwar College of Engineering

(Autonomous Institution)
Chennai - 641 202, Tamil Nadu
Affiliated to Anna University, Chennai



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BONAFIDE CERTIFICATE

Certified that this Report titled **AI-Based Women Safety Smart Slipper** is the bonafide work of

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Submitted for the **Autonomous Semester End Innovative / Multi-Disciplinary Project**

Viva-Voce held on

INTERNAL EXAMINER

EXTERNAL EXAMINER



Sri Eshwar
College of Engineering
Coimbatore | Tamilnadu
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QUALITY POLICY

To establish a system of Quality Enhancement, which would on a continuous basis evaluate and enhance the quality of teaching – learning, research and extension activities of the institution, leading to improvements in all processes, enabling the institution to attain excellence.

INSTITUTE VISION

To be recognized as a premier institution, grooming students into globally acknowledged engineering professionals.

INSTITUTE MISSION

- Providing outcome and value-based engineering education
- Nurturing research and entrepreneurial culture
- Enabling students to be industry ready and fulfill their career aspirations
- Grooming students through behavioral and leadership training programs
- Making students socially responsible

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

DEPARTMENT VISION

To build a conducive academic and research environment in the stream of Artificial Intelligence and Data Science for enabling global education, research and entrepreneurship.

DEPARTMENT MISSION

- Update and enrich the curriculum by integrating global trends and technological advancements to ensure the academic excellence.
- Promote a research-driven environment by providing access to state-of-the-art infrastructure with emerging technologies.
- Empower the faculty through continuous upskilling, international collaborations, and active research engagement to provide globally competent education.
- Encourage student-centric, industry-aligned learning that nurtures innovation and entrepreneurship activities for solving real-world and societal challenges.

PROGRAM EDUCATIONAL OBJECTIVES

Graduates will be able to:

PEO1: Build a progressive professional career in industry, or pursue higher education, or succeed as an entrepreneur.

PEO2: Excel in cutting-edge technologies of Artificial Intelligence and Data Science to address real world challenge

PEO3: Exhibit professional ethics, involvement in team work in their profession and contributing to the advancement of society.

PROGRAM OUTCOMES

PO1 - Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 - Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 - Design/development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 - Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 - Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6 - The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 - Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8 - Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 - Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10 - Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 - Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multi disciplinary environments.

PO12 - Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

PSO1. Apply programming, problem-solving, and hardware integration skills with AI and data science to design intelligent, secure and efficient systems.

PSO2. Develop an AI-driven application by exploring optimized coding competencies using predictive modeling and data analytics technologies.

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ABSTRACT

The proposed project presents an AI-enabled wearable safety system designed to enhance women's security through real-time health monitoring and intelligent emergency response. The system is implemented in the form of a smart slipper, integrating multimodal sensor data and embedded processing to detect potential panic or danger situations automatically.

The device utilizes physiological and motion-based inputs, including heart rate, body temperature, and movement patterns, to analyze the user's condition. These signals are processed using a lightweight AI-based decision model embedded within a microcontroller, enabling the identification of abnormal behavioral patterns such as sudden stress, irregular motion, or distress conditions. Upon detection of a potential threat, the system triggers an automated emergency response mechanism.

The solution incorporates GPS and GSM modules to transmit real-time location data and send SOS alerts to predefined contacts, ensuring rapid assistance without requiring manual user interaction. Additionally, an onboard display and alert system provide immediate feedback and local notification through visual and auditory signals.

The prototype is developed using an embedded systems approach, ensuring portability, low power consumption, and continuous operation as a wearable device. Experimental validation in a controlled environment demonstrates reliable detection capability and real-time responsiveness suitable for practical deployment.

This project highlights the integration of wearable technology, IoT, and artificial intelligence in personal safety applications, contributing toward proactive security solutions, faster emergency response, and improved accessibility in critical situations.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
IoT	Internet of Things
GPS	Global Positioning System
GSM	Global System for Mobile Communication
SOS	Save Our Souls (Emergency Signal)
MCU	Microcontroller Unit
LCD	Liquid Crystal Display
HR	Heart Rate
BPM	Beats Per Minute

CHAPTER 1

INTRODUCTION

The proposed project presents an advanced application of **wearable technology, Internet of Things (IoT), and artificial intelligence** to enhance women's safety through continuous monitoring and real-time emergency response. The system is designed as a smart slipper that functions as an intelligent, non-intrusive safety device capable of detecting panic situations and initiating immediate alerts.

The system integrates multiple sensors to capture physiological and behavioral data, including heart rate, body temperature, and motion patterns. These inputs are processed using an embedded microcontroller and analyzed through an AI-based decision mechanism to identify abnormal conditions such as sudden stress, irregular movement, or distress. By leveraging these multimodal inputs, the system ensures accurate and reliable detection of potential danger scenarios.

Upon identifying a panic condition, the device automatically activates an emergency response system. It utilizes GPS technology to determine the user's real-time location and GSM communication to send SOS alerts and notifications to predefined contacts. Additionally, the system provides immediate feedback through a buzzer and display unit, ensuring both local and remote awareness of the situation.

The architecture of the system is designed to be compact, energy-efficient, and wearable, enabling continuous operation without requiring active user intervention. Unlike conventional safety applications that depend on manual activation, this system emphasizes **hands-free, intelligent safety monitoring**, making it highly effective in critical situations where the user may not be able to access a mobile device.

The primary objective of this project is to develop a reliable and proactive safety solution that enhances personal security, reduces response time during emergencies, and ensures accessibility in real-world scenarios. The system focuses on achieving real-time performance, accuracy in panic detection, and seamless communication for rapid assistance.

CHAPTER 2

LITERATURE REVIEW

Women's safety and personal security systems have become an important research area with the rapid advancement of **wearable technology, Internet of Things (IoT), and artificial intelligence**. The need for proactive and real-time safety solutions has led to the development of various smart devices aimed at improving emergency response and personal protection.

Early safety systems primarily relied on **manual activation mechanisms**, such as mobile applications and panic buttons. These systems allowed users to send emergency alerts and share location details; however, their effectiveness was limited in situations where the user was unable to access the device during distress. Some solutions incorporated basic GPS and GSM modules for location tracking and alert messaging, but they lacked automation and intelligence.

With the emergence of IoT-based systems, researchers introduced **sensor-driven safety devices** capable of monitoring environmental and physical conditions. These systems utilized components such as accelerometers for fall detection and GPS modules for continuous tracking. Although these approaches improved functionality, they were still dependent on predefined thresholds and lacked adaptive decision-making capabilities.

The integration of **machine learning techniques** marked a significant improvement in personal safety systems. Models such as Decision Trees and Support Vector Machines (SVM) were applied to classify abnormal patterns based on sensor data. However, these methods required manual feature extraction and were less effective in handling complex, real-time scenarios involving multiple physiological parameters.

Recent advancements in **artificial intelligence and wearable computing** have enabled the development of more intelligent and autonomous safety systems. Multimodal approaches that combine physiological signals (such as heart rate and body temperature) with motion data provide a more comprehensive understanding of the user's condition. Embedded AI models can now process real-time data to detect stress, panic, or unusual behavior with improved accuracy.

CHAPTER 3

PROBLEM DEFINITION

In today's world, ensuring women's safety has become a critical challenge, particularly in situations such as traveling alone, isolated environments, and emergency conditions. Victims often face difficulty in accessing help quickly due to panic, physical restraint, or inability to use mobile devices. Delays in emergency response and lack of immediate communication can significantly increase risk.

Traditional safety solutions, such as mobile applications and manual alert systems, depend heavily on user interaction. In real-life panic situations, users may not be able to operate their phones or trigger alerts, making these systems less reliable. Hence, there is a need for an **intelligent, automatic, and wearable safety system** that can detect danger conditions and respond instantly without requiring manual effort.

3.1 EXISTING SYSTEM

Existing women safety systems mainly include **mobile applications, wearable panic buttons, and GPS tracking devices**. These systems provide features such as location sharing, emergency alerts, and calling functionalities.

Some IoT-based safety devices use **GPS and GSM modules** to send alerts, while others incorporate **basic sensors** for fall detection or movement tracking. A few systems apply simple threshold-based logic to identify abnormal conditions.

Although these systems offer basic safety features, they largely rely on **manual activation** or predefined conditions and lack intelligent decision-making capabilities.

3.2 DISADVANTAGES OF EXISTING SYSTEM

- Dependence on manual activation during emergencies
- Ineffective when the user is unconscious or unable to respond
- Limited intelligence and lack of real-time decision-making

- High chances of delayed emergency response
- Inability to detect panic or stress conditions automatically
- Limited integration of physiological and behavioral data
- Not fully suitable for continuous, wearable usage

3.3 PROPOSED SYSTEM

The proposed system is an **AI-enabled smart safety slipper** designed to provide continuous monitoring and automatic emergency response. The system integrates multiple sensors to collect physiological and motion-based data such as **heart rate, body temperature, and movement patterns**.

The collected data is processed using a microcontroller and analyzed through an **AI-based decision model** to detect abnormal conditions such as stress, panic, or unusual movement. Unlike existing systems, the proposed solution enables **automatic panic detection without user intervention**.

Upon detecting a potential threat, the system activates an emergency response mechanism. It uses **GPS technology** to determine real-time location and **GSM communication** to send SOS alerts and initiate emergency calls to predefined contacts. Additionally, a buzzer and display unit provide immediate local alerts.

The system is designed as a **wearable, compact, and energy-efficient device**, ensuring continuous operation and ease of use in real-world conditions.

3.4 ADVANTAGES OF PROPOSED SYSTEM

- Automatic panic detection without manual input
- Real-time monitoring and instant emergency response
- Integration of physiological and motion-based data
- Faster communication using GPS and GSM modules
- Continuous wearable safety solution
- Reduced response time during critical situations
- Improved reliability and practical usability

CHAPTER 4

SYSTEM STUDY

System study is a crucial phase in system development that involves analyzing existing solutions, identifying their limitations, and evaluating the feasibility of the proposed system. In this project, the system study focuses on assessing the practicality of developing an **AI-enabled wearable safety device** for women using embedded systems, IoT technologies, and real-time communication modules.

The study ensures that the proposed smart slipper system is **cost-effective, technically implementable, and socially beneficial**, while addressing real-world safety challenges.

4.1 FEASIBILITY STUDY

A feasibility study is conducted to determine whether the proposed system can be successfully implemented from **economic, technical, and social perspectives**. It evaluates resource availability, system performance, and overall impact.

4.1.1 ECONOMICAL FEASIBILITY

The proposed system is economically feasible as it uses **low-cost, readily available hardware components** such as microcontrollers, sensors, and communication modules.

- Components like ESP32, GPS, GSM, and sensors are affordable and widely available
- The overall prototype cost is relatively low compared to its benefits
- No expensive infrastructure is required
- Maintenance and operational costs are minimal

Thus, the system provides a **cost-effective solution for personal safety**, making it accessible to a wide range of users.

4.1.2 TECHNICAL FEASIBILITY

The system is technically feasible due to the availability of **modern embedded systems and IoT technologies**.

- Microcontrollers like ESP32 support real-time data processing
- Sensors enable continuous monitoring of physiological and motion data
- GPS and GSM modules ensure reliable communication and location tracking
- AI-based logic can be implemented using lightweight embedded algorithms

The integration of hardware and software components is achievable, and the system supports **real-time performance, portability, and scalability**.

4.1.3 SOCIAL FEASIBILITY

The proposed system is highly beneficial from a social perspective as it enhances **women's safety and security**.

- Provides immediate assistance during emergency situations
- Reduces response time in critical conditions
- Enables hands-free operation, increasing reliability
- Promotes confidence and independence among users

The system is designed to be **non-intrusive, easy to use, and suitable for daily wear**, ensuring acceptance and usability in real-world scenarios.

CHAPTER 5

SYSTEM REQUIREMENTS SPECIFICATION

The System Requirements Specification (SRS) defines the **hardware and software requirements** necessary to design, develop, and implement the AI-based smart safety slipper. It ensures reliable performance, real-time monitoring, and effective emergency response in practical scenarios.

5.1 HARDWARE REQUIREMENTS

The proposed system requires the following hardware components:

- **Microcontroller:** ESP32 / Arduino (for processing and control)
- **Heart Rate Sensor:** For monitoring pulse rate
- **Temperature Sensor (LM35):** For detecting body temperature
- **Accelerometer (MPU6050):** For motion and fall detection
- **GPS Module (NEO-6M):** For real-time location tracking
- **GSM Module (SIM800L):** For sending SMS and emergency calls
- **LCD Display (16x2):** For displaying system status
- **Buzzer:** For alert indication
- **Power Supply:** Rechargeable battery
- **Voltage Regulator:** For stable power distribution
- **Connecting Components:** PCB, wires, connectors

5.2 SOFTWARE REQUIREMENTS

The system uses embedded programming and communication technologies:

- **Operating System:** Windows / Linux
- **Programming Language:** Embedded C / Arduino C
- **Development Environment:** Arduino IDE
- **AI Logic:** Rule-based / lightweight machine learning model
- **Communication Protocols:** GSM (SMS, Call), GPS data handling
- **Mobile Support:** Android application (for receiving alerts and tracking)

5.3 SOFTWARE DESCRIPTION

The proposed system is developed as an **embedded IoT-based wearable solution** integrating sensing, processing, and communication modules.

The microcontroller acts as the core processing unit, receiving input data from sensors such as heart rate, temperature, and motion sensors. This data is analyzed using an AI-based logic to identify abnormal conditions indicating panic or distress.

Upon detection of a critical situation, the system activates the emergency response mechanism. The GPS module retrieves the user's real-time location, while the GSM module sends SOS alerts and initiates calls to predefined contacts.

The system also includes a local alert mechanism using a buzzer and an LCD display to provide immediate feedback. The overall design ensures **real-time performance, low power consumption, portability, and reliability**.

The integration of hardware and software enables a seamless workflow from **data acquisition** → **processing** → **decision-making** → **emergency communication**, making the system effective for real-world safety applications.

CHAPTER 6

SYSTEM DESIGN

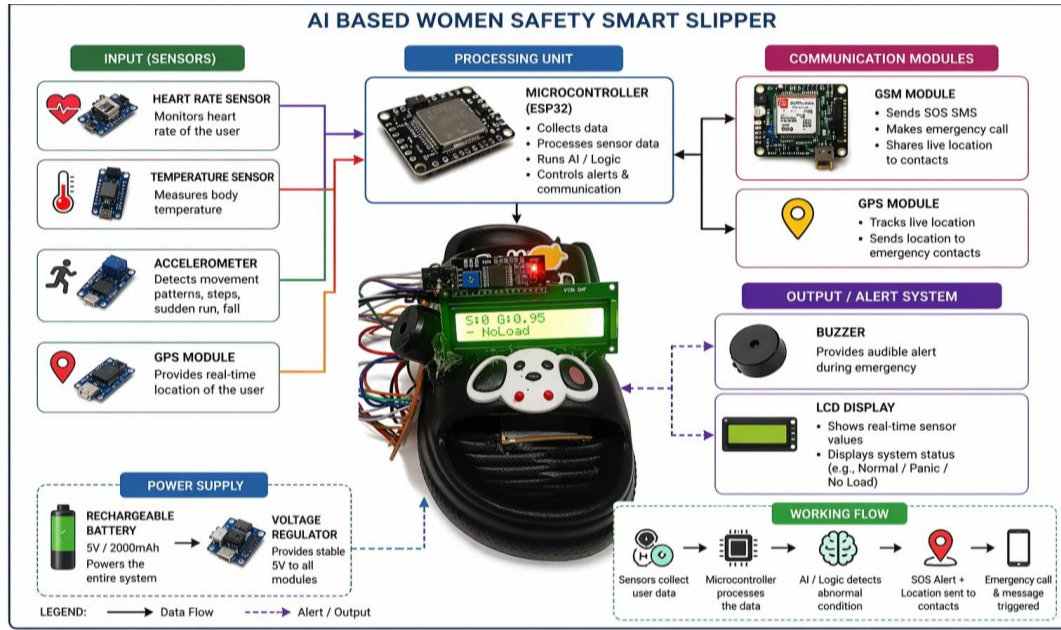


FIG 6.1 SYSTEM ARCHITECTURE

DATA FLOW DIAGRAM (DFD)

AI Based Smart Safety Slipper

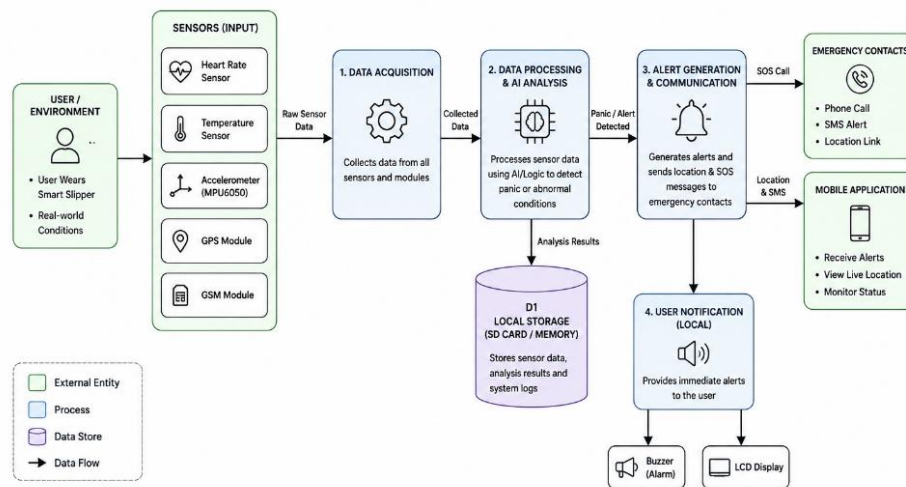


FIG 6.2 DATA FLOW DIAGRAM

CHAPTER 7

SYSTEM IMPLEMENTATION

System implementation is the phase where the designed system is developed and put into operation. In this project, the **AI-Based Smart Safety Slipper** is implemented using embedded systems, IoT modules, and sensor integration. The system continuously monitors user conditions, detects abnormal situations, and provides real-time emergency response efficiently.

7.1 MODULES

The system is divided into the following modules:

- Data Acquisition and Sensor Monitoring
- Signal Processing and Feature Analysis
- AI-Based Panic Detection
- Communication and Alert System
- Real-Time Output and Notification
- Algorithm Implementation

7.2 MODULE DESCRIPTION

Each module plays a crucial role in ensuring the proper functioning of the system.

7.2.1 Data Acquisition and Sensor Monitoring

This module collects real-time data from sensors embedded in the smart slipper.

- Heart rate sensor measures pulse rate
- Temperature sensor records body temperature
- Accelerometer detects movement patterns

The collected data represents the physiological and behavioral state of the user.

7.2.2 Signal Processing and Feature Analysis

In this stage, the raw sensor data is processed to improve accuracy and usability.

- Noise filtering and normalization
- Conversion of analog signals to digital values
- Extraction of meaningful features such as sudden spikes or irregular motion

This ensures reliable input for decision-making.

7.2.3 AI-Based Panic Detection

This module is the core intelligence of the system.

- Analyzes processed sensor data
- Identifies abnormal patterns indicating stress or danger
- Uses rule-based or lightweight AI logic

The system classifies the state as:

- Normal condition
- Panic / abnormal condition

7.2.4 Communication and Alert System

Once a panic condition is detected, this module activates emergency communication.

- GPS module retrieves real-time location
- GSM module sends SOS messages
- Initiates emergency calls to predefined contacts

This ensures quick response during critical situations.

7.2.5 Real-Time Output and Notification

This module provides immediate feedback to the user and surroundings.

- Buzzer generates alert sound
- LCD display shows system status
- Alerts nearby people for assistance

7.2.6 Algorithm

The working of the system can be summarized as follows:

1. Collect data from sensors (heart rate, temperature, motion)
2. Preprocess and normalize the sensor data
3. Analyze data using AI-based logic
4. Detect abnormal or panic condition
5. Retrieve real-time location using GPS
6. Send SOS alert via GSM module
7. Trigger buzzer and display alert message
8. Continue monitoring in real-time

CHAPTER 8

SYSTEM TESTING AND MAINTENANCE

System testing and maintenance are critical phases in the system development lifecycle. Testing ensures that the proposed smart safety system operates correctly, meets user requirements, and performs reliably under different conditions. Maintenance ensures that the system continues to function efficiently after deployment through updates and improvements.

8.1 UNIT TESTING

Unit testing involves testing individual components of the system independently.

- Each hardware component such as heart rate sensor, temperature sensor, accelerometer, GPS, and GSM module is tested separately
- Sensor outputs are verified for accuracy
- Microcontroller functions are tested for correct data processing

This helps in identifying and fixing errors at an early stage.

8.2 INTEGRATION TESTING

Integration testing ensures that all modules work together as a complete system.

- Sensors are integrated with the microcontroller
- Data flow between sensing, processing, and communication modules is verified
- GPS and GSM modules are tested with real-time data transmission

This ensures smooth interaction between all system components.

8.3 FUNCTIONAL TESTING

Functional testing verifies whether the system meets the intended requirements.

- Checks automatic panic detection functionality
- Verifies SOS alert generation
- Ensures correct location sharing via GPS
- Validates buzzer and display response

The system is tested under different simulated conditions such as sudden movement and increased heart rate.

8.4 SYSTEM TESTING

System testing evaluates the complete system performance.

- Testing is done in real-time scenarios (walking, running, sudden stops)
- Checks accuracy of panic detection
- Evaluates response time for alerts
- Ensures reliability and stability of the system

This confirms that the system performs effectively as a wearable safety device.

8.5 WHITE BOX TESTING

White box testing focuses on internal logic and code structure.

- Verifies microcontroller programming logic
- Checks conditions used for panic detection
- Ensures proper execution of loops and decision-making logic

This ensures that the internal working of the system is error-free.

8.6 BLACK BOX TESTING

Black box testing focuses on system functionality without considering internal code.

- Different sensor inputs are provided
- Outputs such as alerts, messages, and buzzer activation are observed
- Ensures correct system response for given conditions

This validates the system from a user's perspective.

SYSTEM MAINTENANCE

After deployment, the system requires regular maintenance to ensure long-term performance and reliability.

Maintenance Activities:

- Fixing hardware and software issues
- Updating AI logic for better panic detection
- Calibrating sensors for improved accuracy
- Enhancing battery performance and power efficiency
- Adding new features such as voice alerts or cloud integration
- Improving overall system performance and reliability

CHAPTER 9

FUTURE ENHANCEMENT AND CONCLUSION

9.1 FUTURE ENHANCEMENT

Although the proposed AI-Based Smart Safety Slipper system performs effectively, several enhancements can be implemented to improve its performance, reliability, and real-world applicability.

- **Improved Detection Accuracy:**
The system can be enhanced by implementing advanced AI/ML models trained on larger and more diverse datasets to improve the accuracy of panic detection under various conditions.
- **Voice-Based Emergency Activation:**
Integration of voice recognition can allow users to trigger SOS alerts using voice commands in situations where manual interaction is not possible.
- **Integration with Emergency Services:**
The system can be connected directly to police stations, hospitals, or emergency response systems for faster and more reliable assistance.
- **Mobile Application Enhancement:**
A dedicated mobile application can be developed with advanced features such as real-time tracking, alert notifications, and emergency history logs.
- **Cloud and IoT Integration:**
Implementing cloud connectivity can enable data storage, remote monitoring, and analytics for better tracking and system improvements.
- **Miniaturization and Wearability:**
Future versions can focus on making the device more compact, lightweight, and comfortable for daily use.
- **Battery Optimization:**
Improving power management techniques can extend battery life and ensure continuous operation.

9.2 CONCLUSION

The AI-Based Smart Safety Slipper system successfully demonstrates the integration of **wearable technology, IoT, and artificial intelligence** for enhancing women's safety. The system is capable of continuously monitoring physiological and movement data, detecting abnormal or panic conditions, and providing real-time emergency response.

By combining multiple sensors with intelligent processing, the system overcomes the limitations of traditional safety solutions that rely on manual activation. The automatic detection and alert mechanism ensures faster response during critical situations, improving the chances of timely assistance.

The developed prototype proves to be **reliable, efficient, and practical for real-world usage**, offering a proactive approach to personal safety. It provides a seamless solution that operates continuously without requiring user intervention.

APPENDIX

A.1.SOURCE CODE

C++

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <TinyGPS++.h>
#include <HardwareSerial.h>

// LCD
LiquidCrystal_I2C lcd(0x27, 16, 2);

// GPS
TinyGPSPlus gps;
HardwareSerial gpsSerial(1);

// GSM
HardwareSerial gsmSerial(2);

// Pins
#define HEART_PIN 34
#define TEMP_PIN 35
#define BUZZER 25

// Thresholds
#define HEART_THRESHOLD 110
#define TEMP_THRESHOLD 38
#define MOTION_THRESHOLD 2000

int lastMotion = 0;

void setup() {
```

```

Serial.begin(115200);

lcd.init();
lcd.backlight();

pinMode(BUZZER, OUTPUT);

gpsSerial.begin(9600, SERIAL_8N1, 16, 17);
gsmSerial.begin(9600, SERIAL_8N1, 26, 27);

lcd.setCursor(0, 0);
lcd.print("Safety System");
delay(2000);
lcd.clear();
}

void loop() {
  // Read sensors
  int heartRate = analogRead(HEART_PIN) / 10; // Simulated BPM
  float temp = analogRead(TEMP_PIN) * (3.3 / 4095.0) * 100;
  int motion = analogRead(33);

  // GPS update
  while (gpsSerial.available()) {
    gps.encode(gpsSerial.read());
  }

  // Display values
  lcd.setCursor(0, 0);
  lcd.print("HR:");
  lcd.print(heartRate);
  lcd.print(" T:");
  lcd.print(temp);

```

```

// Panic detection
bool panic = false;

if (heartRate > HEART_THRESHOLD) panic = true;
if (temp > TEMP_THRESHOLD) panic = true;
if (abs(motion - lastMotion) > MOTION_THRESHOLD) panic = true;

lastMotion = motion;

if (panic) {
    triggerAlert();
}

delay(1000);
}

void triggerAlert() {
    digitalWrite(BUZZER, HIGH);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("PANIC ALERT!");

    String message = "Emergency! Need help. Location: ";

    if (gps.location.isValid()) {
        message += "Lat:";
        message += String(gps.location.lat(), 6);
        message += ",Lng:";
        message += String(gps.location.lng(), 6);
    } else {
        message += "Location not found";
    }
}

```

```

sendSMS(message);

delay(5000);
digitalWrite(BUZZER, LOW);
}

void sendSMS(String msg) {
  gsmSerial.println("AT");
  delay(1000);

  gsmSerial.println("AT+CMGF=1");
  delay(1000);

  gsmSerial.println("AT+CMGS=\"+911234567890\""); // Replace number
  delay(1000);

  gsmSerial.print(msg);
  delay(1000);

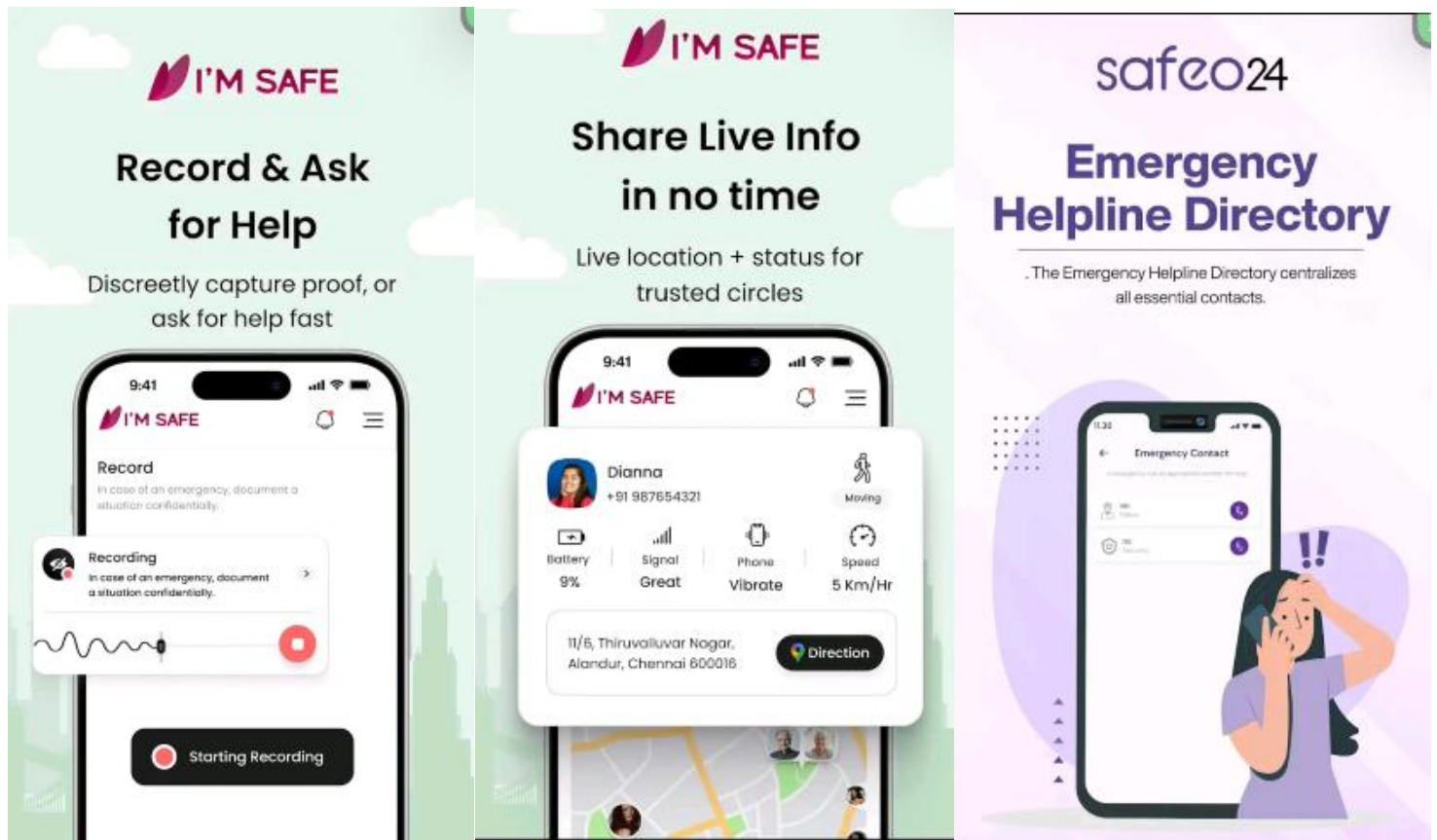
  gsmSerial.write(26); // CTRL+Z to send
  delay(3000);
}

```


6.2.SCREENSHOTS



6.2.1 HARDWARE PROTOTYPE



6.2.2 MOBILE DASHBOARDS

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PROJECT TITLE	AI-Based Women Safety Smart Slipper	
PROGRAM	B.Tech. ARTIFICIAL INTELLIGENCE AND DATA SCIENCE	
PROJECT BATCH NUMBER	1	
BATCH MEMBERS	722823243134	Shruti Kumari Jaiswal
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NAME OF THE SUPERVISOR	Ms.R.Geetha Rajakumari B.E., M.E., (Ph.D)	
NAME OF THE SDG GOALS MAPPED	GOOD HEALTH AND WELL-BEING	
MENTION THE SDG GOALS NUMBER	3	
NAME OF THE TRL LEVEL	Small Scale Prototype	
MENTION THE TRL LEVEL	4	

POs & PSOs Mapping (Put a tick mark in the mapped PO's & PSO's):

Program Outcomes												Program Specific Outcomes	
PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2

Signature of the Supervisor

(Name of the Supervisor)



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VENUE AND EXPENDITURE STATEMENT FOR THE PROJECT WORK

Laboratory details where the project is carried out	Project Lab, ECE
Software / Hardware details	• ESP32 Microcontroller (for data processing and control) • Heart Rate Sensor (for monitoring pulse rate) • Temperature Sensor (LM35) (for body temperature detection) • Accelerometer (MPU6050) (for movement and fall detection) • GPS Module (NEO-6M) (for real-time location tracking) • GSM Module (SIM800L) (for sending SMS and emergency calls) • LCD Display (16x2) (for displaying real-time data)

Details of the Component and Expenditure

S. No	Name of the Component	Qty	Price / Unit in (Rs.)	Amount (Rs.)
1	ESP32 Microcontroller	1	500	500
2	Heart Rate Sensor	1	250	250
3	Temperature Sensor (LM35)	1	300	300
4	Accelerometer (MPU6050)	1	200	200
5	LCD Display (16x2)	1	150	150
6	Buzzer	1	200	200
7	Connecting Wires & PCB	1	250	250

Signature of the student
(Names of the Student)

Signature of the Supervisor
(Name of the Supervisor)